GABRIELA SÁNCHEZ-RODRÍGUEZ

Georgia Tech PhD Candidate ◇ Atlanta, GA
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OBJECTIVE

My research addresses glaucoma as a mechanopathology by developing machine learning algorithms that extract mechanical properties of ocular tissues from OCT retinal imaging, combining finite element simulations with neural networks to enable earlier disease detection.
Research Interests: Physics-informed neural networks, medical image segmentation, computational mechanics, deep learning, retinal biomechanics.

EDUCATION

PhD, Bioengineering (School of Electrical and Computer Engineering), Georgia Institute of Technology
Expected: May 2027
Thesis: Deep Learning and Physics-Informed Neural Networks for Real-Time Estimation of Retinal Mechanical Properties.
Advisor: Dr. Andrew J. Feola
Master of Science in Computer Science (Machine Learning Specialization), Georgia Institute of Technology
2023–2025
Bachelor of Science in Biomedical Engineering, Universidad Carlos III de Madrid
2017–2021
Minor: Advanced Medical Imaging.

AWARDS AND FELLOWSHIPS

RPB Research Travel Award, Emory Eye Center 2026
Awarded Emory Eye Center Travel Award for presenting my research at ARVO 2026.

CRIDC Poster Award, College of Engineering, Georgia Institute of Technology 2026
Awarded at Georgia Tech's internal research poster competition for outstanding presentation.

BrightFocus Foundation Travel Award, BrightFocus Foundation 2021, 2025
Competitive travel award supporting early-career researchers to present at the ISER-BrightFocus Meeting.

Grad Cohort Travel Award, CRA-WP 2022, 2023, 2024
Travel award supporting graduate students to attend the CRA-WP Grad Cohort Workshop.

P.E.O. International Scholar Award, P.E.O. International 2023, 2024
Merit-based award recognizing women of exceptional academic achievement pursuing graduate education in the United States or Canada.

La Caixa Fellowship, La Caixa Foundation, Barcelona, Spain 2022–2024
Highly competitive fellowship supporting top Spanish and Portuguese graduates to pursue doctoral studies at leading universities in the United States and Canada. Awarded by Their Majesties the King and Queen of Spain.

Erasmus+ Scholarship, European Union 2020–2021
European Union scholarship supporting international academic exchange; funded a year abroad studying at FH Technikum Wien.

Academic Excellence Scholarship, Community of Madrid, Spain 2018, 2019, 2020
Merit-based scholarship awarded annually by the regional government of Madrid to undergraduate students with outstanding academic records.

JOURNAL PUBLICATIONS (Google Scholar)

*Equal Contribution, Mentee

[1] (*Under Review*) DeepVFM: Neural Networks Discovery of Virtual Fields for Stable Inverse Constitutive Identification of Hyperelastic Materials.
Sánchez-Rodríguez, G., Sampaio de Oliveira, M. L., Nguyen, T., & Feola, A. J.

[2] (*In preparation*) Ovariectomy modulates gene expression of cell adhesion, cell communication and immune pathways in retinal ganglion cells and glial cells.
Sánchez-Rodríguez, G., Jiang X. ,Bales L. K., Wood, L. B., & Feola, A. J.

[3] (*Under Review*) How Ovariectomy and Aging Affect Vaginal Gene Expression.
Hall, A., **Sánchez-Rodríguez, G.**, Wood, L. B., & Feola, A. J.

[4] (*Preprint, under review*) Forced Exercise Modulates Retinal Inflammatory Response and Regulates miRNA Expression to Promote Retinal Neuroprotection During Degeneration.
Haupt, H., Chen, V. S., Palumaa, T., Anderson, T. E., **Sánchez-Rodríguez, G.**, Chu-Tan, J., et al.
bioRxiv. (2025).

[5] RetOCTNet: Deep Learning–Based Segmentation of OCT Images Following Retinal Ganglion Cell Injury.
Sánchez-Rodríguez, G., Lou, L., Pardue, M. T., & Feola, A. J.
Translational Vision Science & Technology, 14(2), 4. (2025).

[6] Retina-Targeted 17β-Estradiol by the DHED Prodrug Rescues Visual Function and Actuates Neuroprotective Protein Networks After Optic Nerve Crush in a Rat Model of Surgical Menopause.
Prokai-Tatrai, K., Zaman, K., Kapic, A., Hogan, K., **Sánchez-Rodríguez, G.**, Silverio, A. E., Nguyen, V., Prokai, L., & Feola, A. J.
International Journal of Molecular Sciences, 26(5), 1846. (2025).

[7] Ovariectomy Drives Increase of an ECM Transcription Signature in the Posterior Eye and Retina.
Wong, C. A.*, **Sánchez-Rodríguez, G.***, Ethier, C. R., Wood, L. B., & Feola, A. J.
Vision Research, 225, 108507. (2024).

[8] Improved Magnetic Delivery of Cells to the Trabecular Meshwork in Mice.
Bahrani Fard, M. R., Chan, J., **Sánchez-Rodríguez, G.**, Yonk, M., Kuturu, S. R., Read, A. T., Emelianov, S. Y., Kuehn, M. H., & Ethier, C. R.
Experimental Eye Research, 234, 109602. (2023).

[9] AxoNet 2.0: A Deep Learning-Based Tool for Morphometric Analysis of Retinal Ganglion Cell Axons.
Goyal, V., Read, A. T., Ritch, M. D., Hannon, B. G., **Sánchez-Rodríguez, G.**, Brown, D. M., Feola, A. J., Hedberg-Buenz, A., Cull, G. A., Reynaud, J., Garvin, M. K., Anderson, M. G., Burgoyne, C. F., & Ethier, C. R.
Translational Vision Science & Technology, 12(3), 9. (2023).

[10] Transpupillary collagen photocrosslinking for targeted modulation of ocular biomechanics.
Gerberich, B. G., Hannon, B. G., Hejri, A., Winger, E. J., Schrader Echeverri, E., Nichols, L. M., Gersch, H. G., MacLeod, N. A., Gupta, S., Read, A. T., Ritch, M. D., Sridhar, S., Toothman, M. G., Gershon, G. S., Schwaner, S. A., **Sánchez-Rodríguez, G.**, Goyal, V., Toporek, A. M., Feola, A. J., Grossniklaus, H. E., Pardue, M. T., Ethier, C. R., & Prausnitz, M. R.
Biomaterials, 271, 120735. (2021).

CONFERENCE PUBLICATIONS (Google Scholar)

[1] **Sánchez-Rodríguez, G.**, Sampaio de Oliveira, M. L., Nguyen, T. V., & Feola, A. J. DeepVFM: Neural Networks Discovery of Virtual Fields for Stable Inverse Constitutive Identification of Hyperelastic Materials. **WIML Workshop, NeurIPS**, Atlanta, GA. (2026).

[2] Ovariectomy Modulates Transcriptional Signatures Associated with Cell-Adhesion and Immune Pathways in Retinal Ganglion Cells and Glia.
Sánchez-Rodríguez, G., Jiang, X., Bales, K., Wood, L. B., & Feola, A. J.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting). (2026).

[3] MedMoE: Modality-Specialized Mixture of Experts for Medical Vision-Language Understanding.
Chopra, S., **Sánchez-Rodríguez, G.**, Mao, L., Feola, A. J., Li, J., & Kira, Z.

CVPR Workshop on Multimodal Foundation Models for Biomedicine (MMFM-BIOMED), Nashville, TN. (2025).

[4] Deep-Learning Segmentation of the Retina, Choroid, and Sclera from Mice OCT Scans.
Sánchez-Rodríguez, G., Lou, L., Pardue, M. T., & Feola, A. J.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 66(8), 2173. (2025).

[5] Ovariectomy Causes Differential Gene Expression in Retinal Glial Cells.
Sharma, J., Jiang, J., **Sánchez-Rodríguez, G.**, Wong, C., Bales, K., Wood, L., & Feola, A. J.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 66(8), 6314. (2025).

[6] Ovariectomy Modulates Gene Expression of Cell-Adhesion and Immune Pathways in Retinal Glial Cells and Retinal Ganglion Cells.
Sánchez-Rodríguez, G., Jiang, X., Bales, K., Wood, L., & Feola, A. J.
BrightFocus Foundation Glaucoma Research Symposium. (2025).

[7] DeepRetNet 2.0: Deep-Learning Segmentation of the Retina, Choroid, and Sclera from Rat OCT Scans.
Sánchez-Rodríguez, G., Eum, N. C. Y., & Feola, A. J.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 65(7), 2324. (2024).

[8] Treadmill Exercise Decreases Recruitment of Retinal Inflammatory Cells, Alters Retinal miRNA Expression Profiles and Protects Against Retinal Degeneration.
Bales, K., Haupt, H., Palumaa, T., **Sánchez-Rodríguez, G.**, Feola, A. J., Chu-Tan, J., Natoli, R., et al.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 65(7), 1267. (2024).

[9] Age and Ovariectomy Affect Retinal Gene Expression of TGF-β, TNF-α and ECM-Related Pathways.
Sánchez-Rodríguez, G., Wong, C., Wood, L., & Feola, A. J.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 64(8), 406. (2023).

[10] Effects of Age and Ovariectomy on ECM and TGF-β Signaling Gene Sets in Brown Norway Rat Scleras.
Wong, C., **Sánchez-Rodríguez, G.**, Ethier, C. R., Wood, L., & Feola, A. J.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 64(8), 404. (2023).

[11] Deep Learning-Based Segmentation of OCT Images Following Retinal Ganglion Cell (RGC) Injury.
Sánchez-Rodríguez, G., García-Elcano, L., Goyal, V., Pascau, J., Ethier, C. R., & Feola, A. J.
ISER/BrightFocus Glaucoma Symposia: Concepts and Breakthroughs in Glaucoma, Atlanta, GA. (2022).

[12] AxoNet 2.0: A Deep Learning (DL)-Based Tool for Morphometric Analysis of Retinal Ganglion Cell (RGC) Axons.
Goyal, V., **Sánchez-Rodríguez, G.**, Hannon, B. G., Toporek, A. M., Ritch, M. D., Read, A. T., & Ethier, C. R.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 62(8), 1012. (2021).

[13] Deep Learning (DL)-Based Morphometric Analysis of Retinal Ganglion Cell (RGC) Axons in a Rat Model of Glaucoma.
Ethier, C. R., Goyal, V., **Sánchez-Rodríguez, G.**, Brown, D., Hannon, B., Toporek, A. M., Ritch, M. D., Feola, A., & Read, A. T.
Investigative Ophthalmology & Visual Science (ARVO Annual Meeting), 62(8), 2346. (2021).

[14] Reconstruction of Arrays of Acoustic Vortices Using Holograms for Multiple Particle Trapping.
Jiménez, N., **Sánchez-Rodríguez, G.**, Jiménez-Gambín, S., Andrés, D., et al.
The Journal of the Acoustical Society of America, 149(4_Supplement), A18. (2021).

[15] Arrays of Vortices for Particle Trapping and Patterning Using Single-Beam Acoustic Holograms.
Sánchez-Rodríguez, G., Jiménez, N., Jiménez-Gambín, S., & Camarena, F.
51st Spanish Congress of Acoustics (Tecniacústica), Faro, Portugal. (2020).

BOOK CHAPTERS (Google Scholar)

[†]Senior Author

[1] (*In preparation*) Ocular Biomechanics in Menopause and Links with Glaucoma.
Feola, A. J.†, **Sánchez-Rodríguez, G.**, Jiang, D., & Dattilo, M.
In M. J. A. Girard & J. Jonas (Eds.), *What Every Glaucoma Specialist Needs to Know About Biomechanics*. Kugler Publications.

TEACHING EXPERIENCE

Teaching Assistant, BMED 4757: Biofluid Mechanics Jan 2023 – May 2023
Georgia Institute of Technology, Atlanta, GA

- Assisted in teaching core concepts of biofluid mechanics through an inverted classroom format.
- Directed a computational fluid dynamics project simulating Fontan surgical circulation in ANSYS, integrating fluid mechanics principles with clinical applications.

UNDERGRADUATE AND GRADUATE MENTORSHIP

UG: Undergraduate, MS: Master's Student

2023–2026	Aubrey Hall, Biomedical Engineering, Georgia Institute of Technology **[GT President's Undergraduate Research Award ×2]**	UG Student
2023–2025	Alanna Finn, Environmental Studies & Neuroscience, Georgia Institute of Technology **[P&G internship placement]**	UG Student
2024–present	Sanika Mithbaokar, Neuroscience, Emory University	UG Student
2025–2026	Alissa Klein, Biomedical Engineering, Georgia Institute of Technology	UG Student
2025–2026	Leonardo Song, Chemistry, Emory University	UG Student
2025–present	Natalia Celis, Industrial & Systems Engineering, Georgia Institute of Technology	UG Student
2026–present	Meghna Reddi, Neuroscience, Georgia Institute of Technology	UG Student
2026–present	Kai Lewis Yunus, Human Biology & Anthropology, Emory University	UG Student
2026–present	Srishti Iyer, Chemical Engineering, Georgia Institute of Technology	UG Student
2026–2026	Aditya Ghosh, M.S. in Data Analytics, Georgia Institute of Technology	MS Student
2026–present	Meghna Kapa, Neuroscience, Georgia Institute of Technology	UG Student
2026–present	Tatum Simons, Human Biology, Georgia Institute of Technology	UG Student

PEER REVIEW SERVICE

Ad hoc reviewer for the following journals:

- *Translational Vision Science & Technology* (TVST)
- *Ophthalmology Science*

SERVICE AND LEADERSHIP

- Professional Development Chair (2025-2026), Women in Bioengineering Chair (2022-2025) Bioengineering Graduate Association, Georgia Tech. Organized professional events, career panels, and interdisciplinary workshops.
- Founder and Lead, MolVis Media Initiative, Molecular Vision Journal (2026–present). Built and led a global science communication initiative including the EyeSpy podcast and multi-platform social media presence to champion open-access publishing and public engagement with science.
- Co-lead, Genetics Club, Gwinnett Middle Schools (2023–2024). After-school program teaching genetics concepts to middle school students.

TECHNICAL SKILLS

Research Areas	Computer Vision, Deep Learning, Physics-Informed Neural Networks, Bayesian Optimization
Programming	Python, R, MATLAB, JavaScript (basic)
Frameworks	PyTorch, TensorFlow, Scikit-Learn, Weights & Biases
Simulation	FEBio, ANSYS, ABAQUS CAE, SolidWorks

LANGUAGES

Spanish	Native
English	Full professional proficiency
Italian	Conversational fluency
French	Conversational fluency
German	Basic proficiency